

# Zarar Khan

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## HIGHLIGHT OF QUALIFICATIONS

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Junior Full Stack Engineer with experience developing web applications using Python, React, Node.js, Electron and Tailwind CSS. Collaborated with simulation and testing teams to deliver a real-time dashboard for rocket flight metrics, enhancing insights during model testing. Strong problem-solving skills, attention to detail and ability to adapt to new technologies in cross-functional settings.

## SKILLS

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- **Programming Languages & Frameworks:** Python, C++, Visual Studio Code, C#, React, Node.js, Arduino Uno, HTML, Typescript, Javascript, FastAPI, SQLAlchemy, Tailwind CSS, Docker, Verilog, Restful API
- **Tools & Technologies:** Flask, Microsoft Office, Circuit Design, Monogame, SketchUp, Unity, Microsoft Excel, Jupyter Notebook, Electron, Gemini API, AWS
- **Development Skills:** Back-End Development, Front-End Development, Software Development

## EDUCATION

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### University of Waterloo

2030

*Bachelors, Computer Engineering (B.Eng.CE)*

- **Achievements:** Awarded the Presidents Scholarship of Distinction with a +95% admission average

## PROJECTS

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### Hospital Induced Delirium Screening Device | <https://github.com/Zarar3?tab=repositories>

Oct 2025 - Nov 2025

- Created a Screening Device for Hospital Induced Delirium using an Arduino Uno R4 Minima, C++, testing the users cognitively and mechanically, using buttons
- Wrote a Text-To-Speech Python Script for users who may have difficulty reading text
- 3D-Printed a container for the device

### ShadowSync – Hack The Valley | <https://github.com/Zarar3/ShadowSync>

Sep 2025

- Worked in a team of three to create a web application to help users train their form in sports, using Python, React, HTML, JavaScript, Tailwind CSS to create a clean interface for users to use, record and upload videos, including user sign-in and account creation for cybersecurity
- Utilized Gemini's API to create a model that compares a user's form in sports to a professional player, providing a breakdown of their form compared to the professional, giving tips and a similarity score for improvement

## EXPERIENCE

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### University of Waterloo Rocketry Design Team

Sep 2025 - Present

*Firmware Member*

- Collaborated with simulation team to build a real-time dashboard using Python, JavaScript, Node.js, for tracking rocket flight metrics, enabling real-time insights and enhancing decision-making during model testing.
- Authored over 1,000 lines of code to implement new features and automate data processing, demonstrating strong back-end development and software development practices.
- Implemented a board-status display using React, Electron, Vite, and Parsley to present flight statistics with a smooth UI, applying front-end development best practices.